① $P(\text{manchester united win}) = 0.7$

$P(\text{Pub Packed} \mid \text{win}) = 0.9$

$P(\text{Pub Packed} \mid \text{loss}) = 0.6$

$P(\text{manchester united loss}) = 1 - 0.7 = 0.3$

using Bay's theorem:-

$$P(\text{win} \mid \text{Packed}) = \frac{P(\text{Packed} \mid \text{win}) \times P(\text{win})}{P(\text{Packed})}$$

$P(\text{Packed}) = (0.9 \times 0.7) + (0.6 \times 0.3)$

$P(\text{Packed}) = 0.63 + 0.18 = 0.81$

$P(\text{win} \mid \text{Packed}) = (0.9 \times 0.7) / 0.81$

$P(\text{win} \mid \text{Packed}) = 0.63 / 0.81 \approx 0.7778$

$\Rightarrow 77.8\%$ Ans